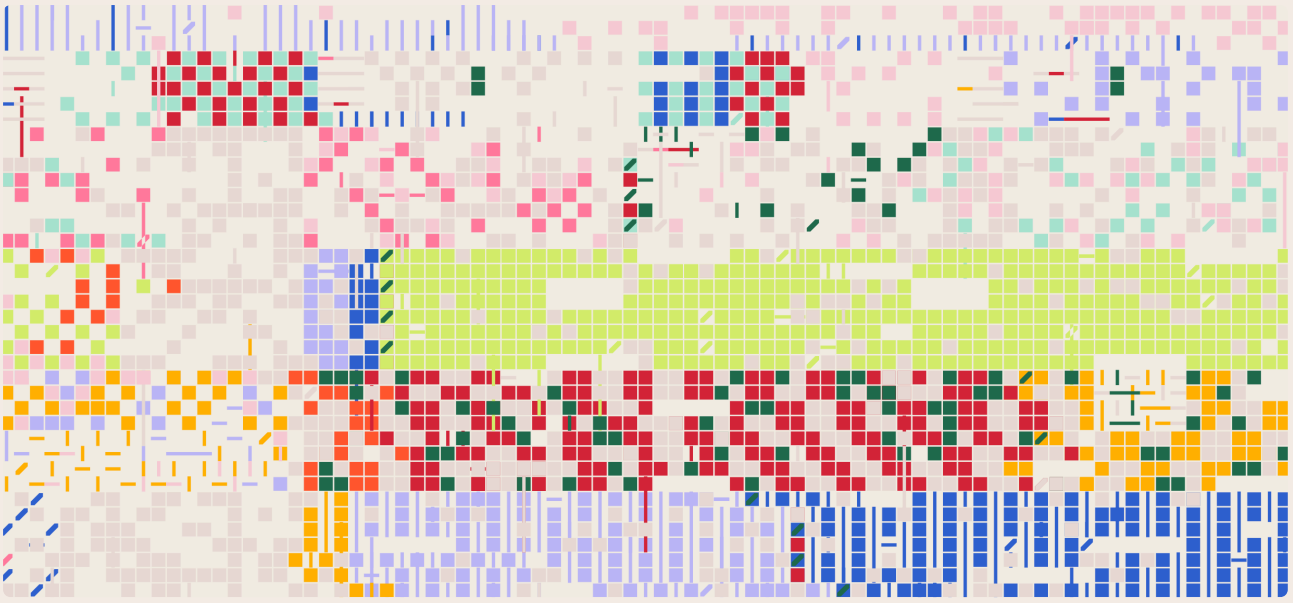


# Espalier

CHORUSES · FEEDBACK · FOLD

created by k.odk

01 · THE IDEA

# A synthesiser with no oscillator

Espalier is an instrument made of four choruses connected in a ring and fed back into themselves, through a wavefolder and a resonant band-pass filter.

There is nothing to play back and nothing to trigger. The sound is the loop itself: the choruses hear each other, and when the feedback is strong enough the ring begins to sing. Espalier holds the loop at the edge where it only just rings — it creeps towards ringing, backs off, and creeps again — and you play that edge with your fingers.

There is no LFO and nothing random. What moves the sound are four envelopes, in the manner of a modular envelope generator, that listen to the loop and to each other. You can patch them into one another with cables, right on the screen.

**The screen is the instrument.** It is one printed sheet of small cells, with no words. Each area of cells is a part of the instrument and draws what that part is doing: the weaves drift, swell and glitch with the sound. You play it by touching it.

**Too much to learn? Skip it.** The sound starts by itself when the app opens. Put one finger on the lower right of the screen and drag slowly — sideways and up and down — and listen. Then double-tap there: the loop is wired a different way, and the whole sheet changes colour. That is most of it.

**Contents**


02 Requirements

03 The screen at a glance

04 The bands — the four choruses

05 The left rows — the whole loop

06 The envelopes

07 Cables

08 All gestures on one page

09 Support

# What it needs

Espalier runs the whole ring at audio rate, sample by sample, while it draws the sheet thirty times a second.

### System

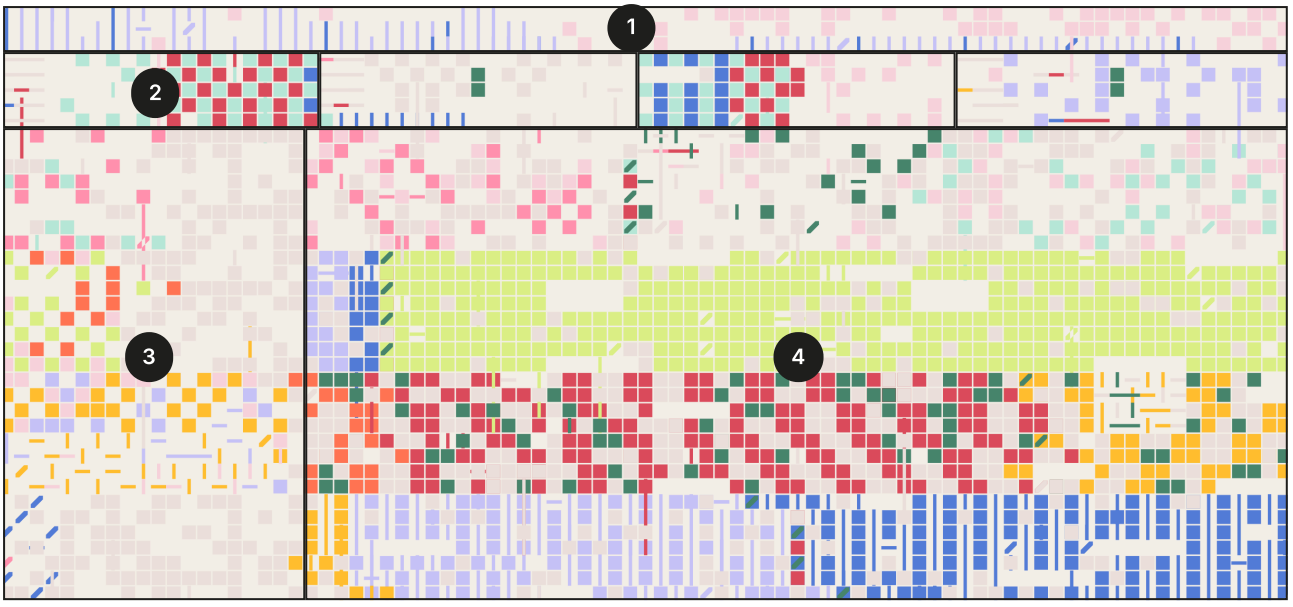
| | |
|---|---|
| IOS | iOS 18.0 or later. |
| DEVICE | iPhone, played in landscape. The whole screen is the instrument; there are no menus or panels. |
| RECOMMENDED | iPhone 13 or newer. Espalier is developed and tested on an iPhone 13. |
| BACKGROUND | It keeps playing while you switch apps or lock the screen, the way an instrument should — so you can play it alongside other music apps or record it elsewhere. Stop it from the lock screen or Control Centre, or close the app. |
| MICROPHONE | Not used. Nothing is recorded. |
| PRIVACY | No account. Nothing is collected and nothing leaves the device. |

### Starting

Open the app. There is no start button: the loop begins on its own and swells within a few seconds. If you hear nothing, check the silent switch and the volume, then move a finger on the lower right.

# One sheet, four areas

The sheet is 85 cells across and 39 down. It is divided into four areas, and what a finger lands on is what it plays.




The sheet in its first wiring (series). Each wiring has its own inks — see page 4.

| | |
|---|---|
| 1 | **The wave** — the thin bars along the very top. They are the sound of the loop, drawn as it happens. Double-tap here to put every cable back where it started. |
| 2 | **The envelopes** — four strips across the width, one envelope each. They push the rest of the instrument around. Page 6. |
| 3 | **The left rows** — four rows on the left, one above the other. Each moves two settings of the whole loop. Page 5. |
| 4 | **The bands** — the four wide rows on the right are the four choruses, top to bottom. Page 4. |

The rows of the left side and the bands on the right line up: the first left row sits beside the first band, and so on. A row and its band share one envelope.

# The four choruses

Each of the four wide bands on the right is one chorus in the ring. This is where you play most of the time.

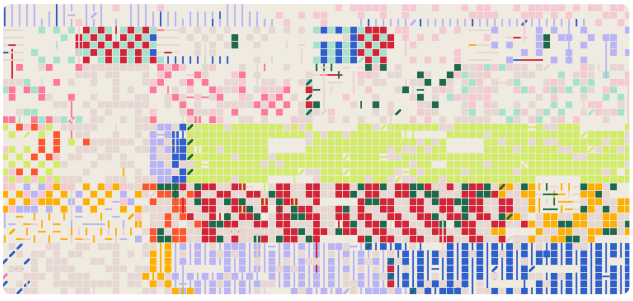
### One finger on a band

| | |
|---|---|
| ACROSS | The chorus's **delay**. Left is short, right is long. The seam in the band's weave shows where it is. |
| UP AND DOWN | How much of the chorus is **mixed back** into the ring. Higher is thicker; the weave gets denser. |

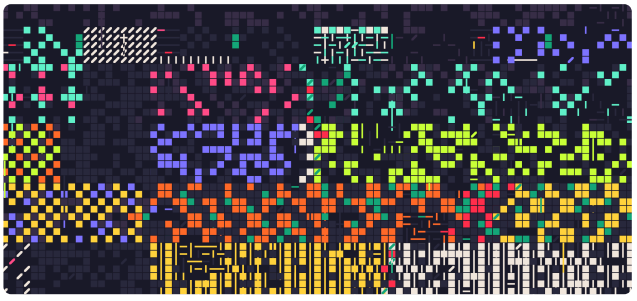
Small moves matter most near the edge: the ring can jump from silence to singing with a few millimetres. Where the band's envelope is pushing it, a third ink spills out of the seam.

### Double-tap a band — the wiring

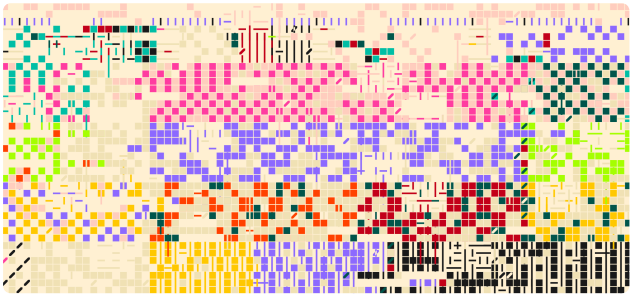
The four choruses are wired into the ring a different way, and the whole sheet changes ink. There are five wirings, in turn: **series**, **upside down**, **crossed**, **folded** and **nested**. The same finger positions sound quite different in each.



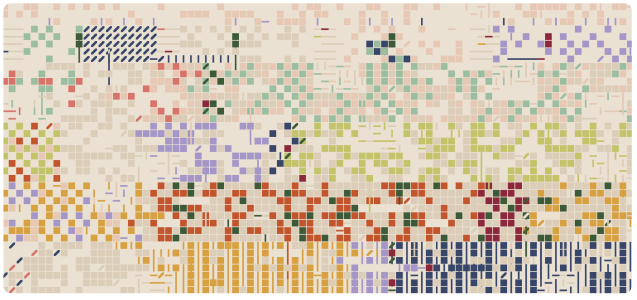
Series — warm paper



Upside down — at night



Folded — acid



Nested — clay

(Crossed, in sea blue and rust, is between upside down and folded.)

05 · THE LEFT ROWS

# The whole loop

The four rows on the left each move two settings that act on the whole ring, not on one chorus.

### Drag on a row

| | |
|---|---|
| ROW 1 · FILTER | The resonant band-pass at the end of the ring. Across: its centre frequency. Up: its resonance (Q). |
| ROW 2 · FOLD | The wavefolder in the ring. Across: how deep it folds. Up: how hard it is driven. |
| ROW 3 · EDGE | How close to ringing the loop is held. Across: how loud the edge rings. Up: how far past the edge it goes before it turns back — low holds it right on the edge, high lets it swell out and fall silent. |
| ROW 4 · WHOLE | Across: the length of every delay together (scaling all four choruses). Up: brightness. |

The row is woven as far across as the first setting, and its two inks are mixed as much as the second.

### Tap a row — what moves that chorus

Each row also belongs to the chorus beside it. A tap steps through what moves that chorus, shown by the row's pattern:

| | |
|---|---|
| CHECKS | Your hand only. |
| CROSSES | The loop's own loudness. |
| SLASHES | The sound coming into that chorus. |

# Four envelopes in a ring

The four strips across the top are four envelopes. Each one pushes its row and its chorus around. They listen to the loop, and to each other: each hears the one before it, and the first hears the last.

### On a strip

| | |
|---|---|
| DRAG | How far and in which direction the envelope pushes. The cells grow from the dark stem in the middle towards your finger; crimson shows where the envelope is right now. |
| DOUBLE-TAP | The envelope's mode, in turn. **Follow** (thin bars): it follows the loop's loudness. **Sync** (slashes): one rise and fall each time the envelope before it peaks. **Free** (checks): it cycles over and over, faster the higher the one before it is. **Chain** (lines): it follows the one before, at its own pace. |
| TWO FINGERS ON ONE STRIP | Tap with a second finger while the first holds: the envelope becomes **independent** — no cable reaches it. Its ground turns to green lines. Again to undo. |
| HOLD ONE, TAP ANOTHER | The two envelopes play **ping-pong**, each firing the other. Again to stop. |
| HOLD STILL | Keep a finger still on a strip for most of a second: every cable going into it is pulled out. |

### How fast — two fingers

Put one finger on a band or a left row, and a second finger anywhere. The distance between them sets that row's envelope: how far apart **sideways** is the rise, how far apart **up and down** is the fall. Close together is fast — down to audio rate, where the envelope becomes a tone of its own. Wide apart is slow, up to ten seconds.

When the loop rings loud, an envelope may be thrown into cycling by itself (it turns to checks) — the louder, the faster — and falls back as the loop quietens. This is part of how Espalier keeps itself at the edge.

# Patching on the screen

The envelopes can be patched into each other, as on a modular synthesiser — by drawing a cable with your finger.

## Drawing a cable

Start on the **stem** of a strip (the narrow dark column in its middle) and drag. A line follows your finger. Let go over another strip to plug in. Where you let go decides what the cable does:

| | |
|---|---|
| UPPER HALF, LEFT THIRD | End of cycle fires the other envelope. |
| UPPER HALF, MIDDLE | End of rise fires the other envelope. |
| UPPER HALF, RIGHT THIRD | The output pushes both its rise and its fall. |
| LOWER HALF, LEFT THIRD | The output pushes its rise. |
| LOWER HALF, MIDDLE | The output pushes its fall. |
| LOWER HALF, RIGHT THIRD | The output, upside down, pushes both. |

Draw the same cable again to unplug it. The three columns at the left edge of a strip show what is plugged into it, one row per source, in that source's ink (crimson when upside down). The stem turns crimson while the end of rise is high.

## Sending an envelope elsewhere

Let a cable go anywhere outside the strips and the envelope is sent there as well: onto a left row (its two settings, or that chorus's depth), onto a band (that chorus's delay or mix), or onto the wave (filter, fold or length). A thin column in the envelope's ink marks each place, as tall as the envelope is now. Each envelope can be sent to two more places.

## Starting over

Double-tap the wave along the very top: every cable goes back to how it started.

# On one page

Everything a finger can do, by where it lands.

### Bands — right

| | |
|---|---|
| DRAG | That chorus: delay across, mix up. |
| DOUBLE-TAP | The next wiring of the ring; the sheet changes ink. |
| + SECOND FINGER ANYWHERE | That row's envelope: rise across, fall up and down. |

### Left rows

| | |
|---|---|
| DRAG | Two settings of the whole loop (filter · fold · edge · whole). |
| TAP | What moves that chorus: hand · loudness · sound coming in. |
| + SECOND FINGER ANYWHERE | That row's envelope: rise across, fall up and down. |

### Envelope strips — top

| | |
|---|---|
| DRAG | How far the envelope pushes. |
| DOUBLE-TAP | Mode: follow · sync · free · chain. |
| DRAG FROM THE STEM | A cable: to a strip, a patch; elsewhere, one more place it pushes. |
| HOLD STILL | Pull out every cable going into it. |
| SECOND FINGER, SAME STRIP | Independent (on / off). |
| HOLD ONE, TAP ANOTHER | Ping-pong (on / off). |

### The wave — very top

| | |
|---|---|
| DOUBLE-TAP | Every cable back to the start. |

# Help and contact

Espalier is made by k.odk, a musician in Okayama, Japan.

| | |
|---|---|
| SUPPORT PAGE | keiodak.github.io/privacy/espalier-support.html |
| PRIVACY | keiodak.github.io/privacy/espalier.html — nothing is collected. |
| CONTACT | kodk.dev@gmail.com — please include your device model, iOS version and what you were doing. |

## Common questions

| | |
|---|---|
| IT GOES QUIET | That is the edge at work: the loop backs off and comes back. Give it a moment, or move a band upwards. |
| IT IS TOO WILD | Move row 3 (edge) downwards to hold it on the edge, or double-tap the wave to put the cables back. |
| IT KEEPS PLAYING | It plays in the background on purpose. Stop it from the lock screen or Control Centre. |
| RECORDING | Use the iOS screen recorder, or record the output through an audio interface. |